High-Resolution Remote Sensing Image Change Detection Based on Fourier Feature Interaction and Multiscale Perception

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***Abstract*— As a significant means of Earth observation, change detection in high-resolution remote sensing images has received extensive attention. Nevertheless, the variability in imaging conditions introduces style discrepancies and a range of pseudochange regions between bitemporal image pairs. Furthermore, changing objects possess diverse morphological representations, which makes accurately identifying change areas and delineating their boundaries within complex object distributions increasingly difficult. In response to the aforementioned challenges, we propose the Fourier feature interaction and multiscale perception (FIMP) model for effective change detection. To mitigate the impact of style discrepancies, FIMP employs the Fourier transform to adaptively filter bitemporal features in the frequency domain while mining the optimized bitemporal features relevant to the change detection task. To enhance the ability to recognize multiscale changing objects, FIMP aggregates and emphasizes the change areas with the introduced temporal change enhancement module (TCEM). By utilizing the U-fusion change perception module (UCPM) to perform multilevel bidirectional fusion of change features at different scales, FIMP can further enhance the ability to delineate complex semantic change boundaries. Experiments on three public datasets show that our approach outperforms seven state-of-the-art methods.**

***Index Terms*— Change detection, Fourier feature interaction, high-resolution remote sensing image, multiscale change feature.**

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I. Introduction

CHANGE detection identifies surface alterations of the Earth by analyzing remote sensing images acquired at different times but from the same geographical location [1], [2]. These changes encompass a variety of phenomena, such as the construction of new buildings, the repair of roads, the expansion of agriculture, the degradation of forests, and environmental pollution. Therefore, change detection has found wide applications in various fields, including urban planning [3], disaster assessment [4], and natural resource management [5]. With the rapid advancement of remote sensing technology, the accessibility of high-resolution remote sensing images is increasing, further driving the progress of change detection techniques.

Early traditional change detection methods can primarily be categorized into algebra-based approaches and transformation-based approaches [6]. Algebra-based methods obtain change detection results for each pixel through algebraic or statistical means, such as change vector analysis (CVA) [7] and segment anything model (SAM) [8]. Transformation-based methods convert the original images into other feature spaces to distinguish changed and unchanged areas, such as principal component analysis (PCA) [9], independent component analysis (ICA) [10], and linear discriminant analysis (LDA) [11]. However, when faced with complex land cover distributions, traditional methods frequently exhibit limited detection performance. This limitation arises from their dependence on handcrafted features, which compromises robustness.

In recent years, with the widespread adoption of deep learning, an increasing number of researchers have begun to apply deep learning to change detection tasks [12]. Due to the powerful feature learning capabilities and hierarchical learning structures of convolutional neural networks (CNNs), change detection technology has been elevated to a new level. Fully convolutional early fusion (FC-EF) [13] adopts an early fusion method that directly concatenates bitemporal images along the channel dimension as new input, performing end-to-end change detection tasks within a single stream framework. Nonetheless, it overlooks the consideration of temporal correlation, leading to an inability to accurately locate the changed areas. Compared to the single-stream architecture, the dual-stream framework utilizes Siamese networks to extract multilevel abstract features from bitemporal images [6]. It is more adept at learning and distinguishing the similarities

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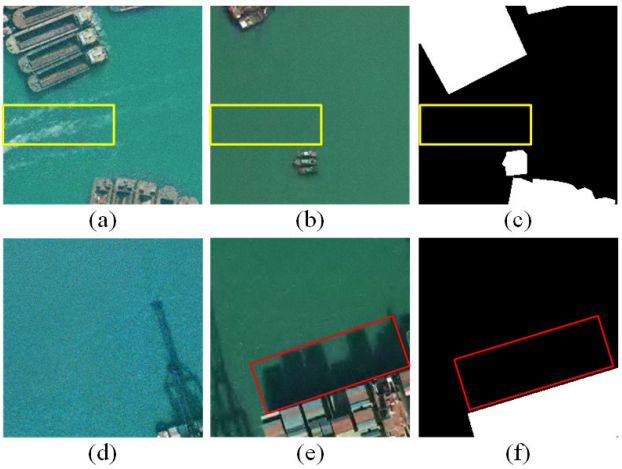


Fig. 1. Inconsistency in imaging conditions leads to stylistic discrepancies in bitemporal images (yellow boxes), as well as building shadows caused by variations in solar elevation angles (red boxes). (a) and (d) T1 image. (b) and (e) T2 image. (c) and (f) Ground truth.

or differences between bitemporal instances. Consequently, subsequent researchers have shown a preference for adopting the dual-stream architecture to accomplish change detection tasks, such as [14], [15], [16], and [17]. With the emergence of attention mechanisms, many researchers have incorporated them into change detection models to achieve better feature representation. Attention mechanisms improve the efficiency and performance of change detection by computing weights for feature maps to capture key information relevant to the task, such as [18], [19], [20], [21], and [22]. However, due to the limitation of the fixed receptive field of convolutional kernels, the aforementioned methods struggle to capture the contextual information of the input images. To better capture long-range information and overcome the limitations of the fixed receptive field of convolutional kernels, many researchers have further introduced transformers into change detection models [23], [24].

However, the methods mentioned primarily concentrate on enhancing feature extraction from bitemporal images and often overlook the unique characteristics of change detection tasks. Specifically, given the variability in imaging environments, maintaining consistent imaging conditions at the same geographical location over different times is exceedingly challenging for sensors. Therefore, inevitably, there are pseudochanges, which manifest as inconsistencies in the representation of areas that remain unchanged in bitemporal images. These discrepancies can include shadows, brightness, and stylistic discrepancies between the images. As highlighted in the yellow boxes in Fig. 1(a)–(c), the color and brightness of these regions within the bitemporal images are noticeably inconsistent. In addition, there are waves or wakes present in nonchanging areas. The red boxes in Fig. 1(d)–(f) highlight the shadows cast by buildings due to variations in solar elevation angles during imaging. The presence of such pseudochanges can interfere with the effective extraction of change features by the model, leading to misinterpretations of change areas.

In view of the above problems, some researchers employ image translation to adjust the styles of bitemporal images. This approach explicitly reduces the impact of pseudochanges resulting from inconsistencies in style between the images [25], [26], [27]. However, the detection performance of the model largely depends on the quality of image translation. Therefore, some other researchers have started to adopt a domain adaptation approach to implicitly mitigate the impact of pseudochanges [28], [29]. Nevertheless, the aforementioned methods utilize multitask learning or generative adversarial approaches to align the feature distributions of bitemporal images, significantly increasing the complexity of model training.

Furthermore, in change detection tasks, the changing objects exhibit complex and diverse morphological representations. The diversity here is intuitively reflected in the varying sizes and irregular shapes of changing ground objects, which makes it more challenging to simultaneously recognize changing objects of different scales and accurately delineate their change boundaries.

To address the two major challenges, a change detection method for high-resolution remote sensing images based on Fourier feature interaction and multiscale perception (FIMP) is proposed. First, FIMP mitigates the inevitable style differences in change detection tasks from a frequency-domain perspective. Considering that the change of a single element in the frequency domain will affect the global distribution of the original features [30]. FIMP filters the bitemporal frequency-domain features according to the task discrimination information contained in different frequency components to optimize the feature representation, thereby mitigating the influence of pseudochanges such as style discrepancies. Furthermore, to enhance the model's ability to accurately identify changing objects in complex scenarios, FIMP models the temporal correlation of bitemporal images using the temporal change enhancement module (TCEM) to capture change areas. Subsequently, a U-fusion change perception module (UCPM) is introduced to aggregate the rich contextual information between multiscale features [31]. This enhances the model's ability to perceive the complex spatial forms of change objects, achieving precise localization of multiscale change boundaries and change objects.

The main contributions can be summarized as follows.

1) To optimize the global representation of bitemporal features with frequency-domain features, a Fourier feature interaction strategy is proposed, with a designed frequency-domain feature interaction framework. This framework employs an adaptive frequency filtering module (AFFM) to adaptively weight different frequency components within bitemporal features, which can not only optimize the global representation of features but also mine frequency components relevant to downstream tasks.
2) To fully model the temporal correlation of bitemporal images and obtain change features, a TCEM is proposed. TCEM captures more representative change information and highlights change regions through the full interaction with bitemporal features. It provides preliminary guidance for the network to perceive multiscale changing objects.
3) To aggregate the rich contextual information between multiscale change features, a UCPM is introduced. UCPM narrows the semantic gap between features of

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various scales through multilayer bidirectional aggregation, enhancing the network's ability to perceive complex changing objects.

The remainder of this article is organized as follows. Section II reviews some change detection methods based on deep learning and the Fourier transform of visual representation learning. Section III introduces the designed FIMP in detail. Section IV presents and analyzes the experimental results. Finally, Section V concludes this work with summarized future directions.

II. Related Work

A. Deep Learning-Based Change Detection Methods

In recent years, change detection methods based on deep learning have shown great vitality. Daudt et al. [13] introduced the Siamese network into change detection, leveraging the shared-weight encoder to enhance the ability to detect changes. This dual-stream architecture facilitates the network's learning and differentiation between bitemporal instances, which has led to its increasing adoption by researchers for change detection tasks. Zhao et al. [14] designed a novel dual-stream change detection framework using a semantic-guided strategy to overcome the interference issues present in bitemporal feature fusion. Liu et al. [15], based on a dual-stream framework, achieved better detection results by aggregating spatial detail information and spectral difference information. Chen et al. [16] employed scale-invariant learning and local interaction to achieve change detection on continuous cross-resolution remote sensing images.

With the advent of attention mechanisms, many researchers have incorporated them into change detection models to enhance the focus on task-relevant information [19], [21]. Lv et al. [32] employed a spatial-spectral attention mechanism to capture more representative change features. Xu et al. [22] employed an attention pyramid and channel-cross attention mechanism to focus on crucial semantic information within channels and depth information. As a representative global self-attention method, a transformer has also achieved impressive results in change detection tasks [23]. Chen et al. [18] employed a transformer to capture long-range information in bitemporal images, improving the detection of change boundaries and small targets. Jiang et al. [24] utilized a transformer to learn a consistent representation of the background portions of bitemporal images, enhancing the ability to recognize change areas.

In addition, several methods have been proposed to enhance the performance of change detection tasks by incorporating advanced models [33]. Mei et al. [34] leveraged the powerful visual recognition capabilities of the SAM to better extract and integrate contextual semantics, thereby improving the accuracy and robustness of semantic change detection. Dong et al. [35] explored the potential of multimodal data in the field of change detection by reconstructing contrastive language-image pretraining (CLIP) to extract bitemporal features.

B. FT for Visual Representation Learning

In visual representation learning, Fourier transformation (FT) [36] is gaining attention as a valuable tool for frequency-domain analysis. It extracts frequency information from images, enhancing models' understanding of global and local features. Some researchers have improved model generalization by processing images in the Fourier frequency domain, perturbing or enhancing components within specific frequency ranges [37]. FSDR [38] decomposes images into multiple frequency components, achieving domain generalization in semantic segmentation tasks by retaining domain-invariant components and randomizing domain-variant components. ALOFT [39] dynamically perturbs the low-frequency components of images, disrupting local textures while preserving global structural features, in order to reduce model overfitting to the source domain. In addition, some researchers have explored incorporating FT into the network training process, combining frequency-domain features with time-domain features to enhance the model's learning capability [40]. FsaNet [41] designed a low-frequency self-attention module that better preserves edge information in semantic segmentation tasks. Liu et al. [30] fully exploited semantic information and fine-grained features by enhancing low-frequency features in the encoder and high-frequency features in the decoder.

Several researchers have integrated frequency-domain analysis techniques into the domain of remote sensing image change detection [42]. Zheng et al. [43] designed a high-frequency-guided Siamese network, utilizing a high-frequency attention module to enhance the network's focus on high-frequency information related to buildings. Tang et al. [44] employed wavelet transform to decompose features into different frequency components for separate interactions, completing object fine-grained change detection task.

The aforementioned methods each focus on different aspects of the frequency domain, but they do not utilize frequency-domain information to address style discrepancies between bitemporal images. In this article, we transform the bitemporal features into the frequency domain using FT and apply adaptive frequency filtering to facilitate interaction between the bitemporal images, thereby mitigating the impact of style differences to some extent.

III. Proposed Method

In this section, the overall framework of the proposed method is introduced. The Fourier feature interaction strategy employed by FIMP is detailed, followed by introductions to AFFM, TCEM, and UCPM. Finally, the loss functions used are discussed.

A. Overview

As shown in Fig. 2, the proposed FIMP mainly consists of a backbone, the Fourier feature interaction strategy (implemented by the AFFM), the TCEM, the UCPM, and a decoder. The first five layers of EfficientNet-b4, equipped with pretrained weights, are used as the backbone network. Thanks to the effectiveness of the compound scaling strategy

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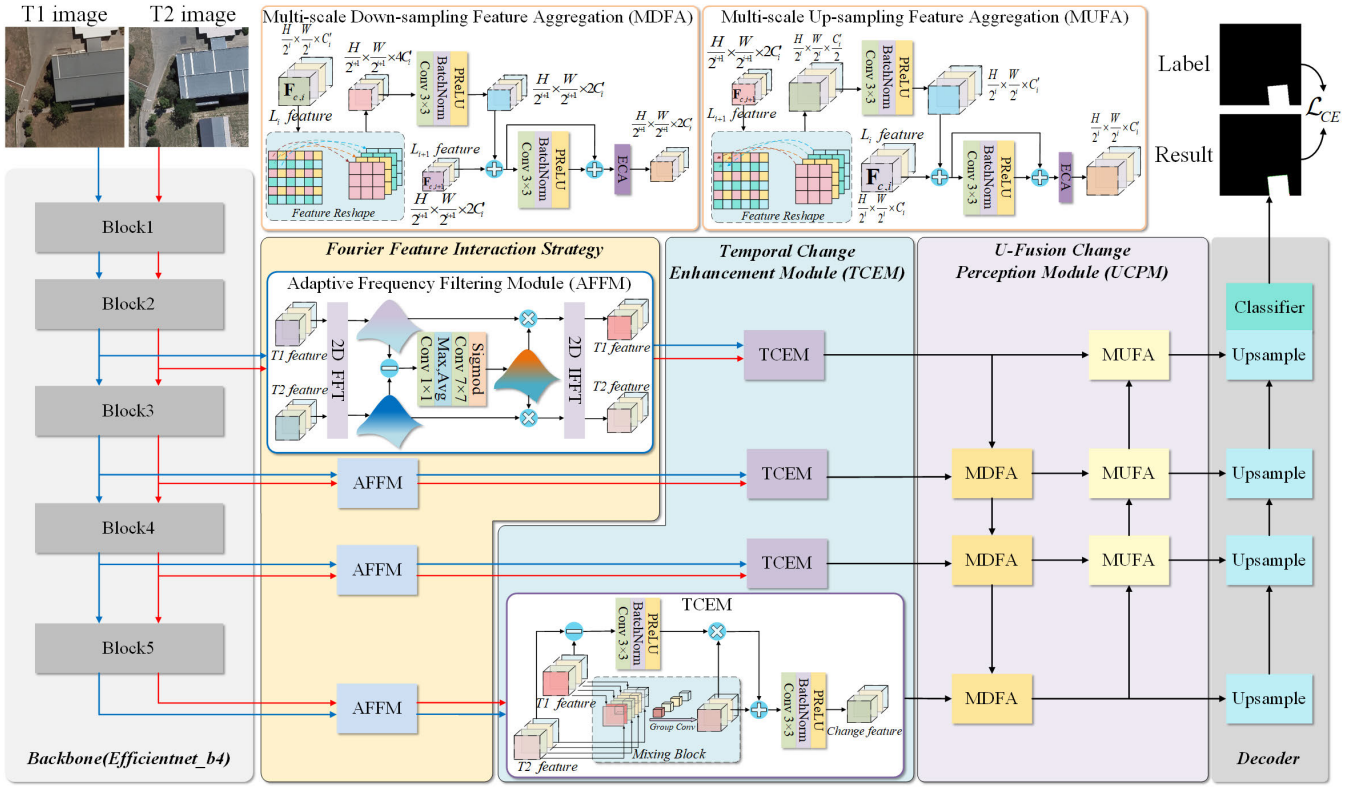


Fig. 2. Framework of FIMP, which is mainly composed of five parts: a pretrained backbone, Fourier feature interaction strategy implemented by the AFFM, TCEM, UCPM, and a decoder. In FIMP, blocks 1–5 represent the first five layers of EfficientNet-b4, which with pretrained weights. The bitemporal features are updated for global representation through AFFM. Multiscale change features are obtained via TCEM, and the multiscale feature semantics are aggregated using UCPM. Ultimately, the features are fed into the decoder to produce the prediction results.

and the depthwise separable convolutions, EfficientNet has a lower computational cost and competitive performance [45]. Therefore, it is highly suitable as a feature extractor in change detection tasks to obtain multiscale feature maps from bitemporal images.

Let a pair of bitemporal images be $\{\mathbf{I}^1, \mathbf{I}^2\} \in \mathbb{R}^{H\times W\times 3}$, where H represents the height of the image, W represents the width of the image, and 3 represents the band of the image. After feeding the bitemporal images into the backbone, multiscale bitemporal feature maps $\{\mathbf{F}_i^1, \mathbf{F}_i^2\} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i}$ are extracted, where $i \in \{0, 1, 2, 3, 4\}$ represents the ith layer of the backbone, and $(H/2^i)$, $(W/2^i)$, and C_i represent the height, width, and channel of the ith layer features, respectively. Subsequently, through the Fourier feature interaction strategy, the frequency features of bitemporal images are optimized by using AFFM. By mining different frequency components in the feature map, the frequency components related to the change detection task are enhanced, and the influence of redundant information is reduced. After adaptive frequency filtering, a TCEM is utilized. This module, operating in a manner similar to differential attention, fully interacts with bitemporal features to capture change information across different scales of change objects.

To equip the model with the capability to capture the complex and diverse spatial morphological representations of change objects, UCPM is employed to aggregate feature maps across different scales. An architecture similar to U-Net is utilized to fully integrate multiscale features, enabling the precise identification of multiscale change targets while accurately locating complex change boundaries. Finally, the final change detection result map is obtained through layer-by-layer upsampling and a classifier.

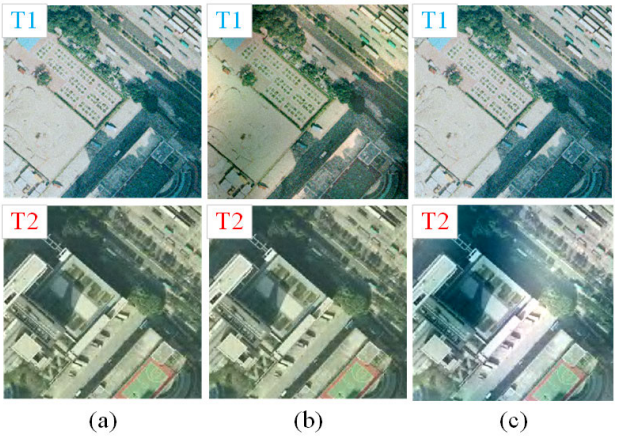


Fig. 3. After performing the Fourier transform on the bitemporal images, replace their low-frequency components. The colors and brightness of the bitemporal images are also replaced accordingly. (a) Original bitemporal images. (b) Replacing the low-frequency components of the T1 image with those of the T2 image. (c) Replacing the low-frequency components of the T2 image with those of the T1 image.

B. Fourier Feature Interaction Strategy

For a given image, the low-frequency information is related to the slowly changing gray-scale components, such as the image's color and brightness [38], [46], [47], [48]. The high-frequency information is related to the rapidly changing

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gray-scale components, typically manifesting as the object's edge contours and speckle noise [30]. As shown in Fig. 3, after the fast Fourier transform (FFT) [36] of the bitemporal images, the low-frequency component of the T1 image is replaced with the T2 image, and then, the T2 image is reconstructed by inverse FFT (IFFT). It can be observed that the color and brightness of the T2 image have changed and become similar to the T1 image, and vice versa. This further illustrates the correlation between frequency components and style representation. This characteristic aids in suppressing the impact of pseudochanges caused by style discrepancies between bitemporal images through the use of frequency-domain information.

During 2-D FFT [36] of an image, different frequency components in the image are decomposed into elements at different positions. This means that the entire image can be represented as a superposition of multiple frequencies, with each frequency corresponding to different structures and details in the image. Furthermore, in the frequency-domain representation of an image, updating individual frequency-domain elements affects the global representation of the entire image [49]. Based on this characteristic, a Fourier feature interaction strategy is proposed.

First, the 2-D FFT [36] is applied to the spatial dimension of bitemporal features $\{\mathbf{F}_i^1, \mathbf{F}_i^2\}$ after the backbone, transforming the features into the frequency domain. Since the features after FFT are in complex form, for ease of calculation, the real and imaginary parts are stacked along the channel dimension to obtain $\mathbf{F}_{\mathrm{FFT},i}^{1/2}$, which can be formulated as

$$\begin{aligned}\mathbf{F}_{\mathrm{FFT},i}^{1/2}(u,v) &= \mathcal{F}\left[\mathbf{F}_i^{1/2}(h,w)\right] \\ &= \sum_{h=1}^{H-1}\sum_{w=1}^{W-1}\mathbf{F}_i^{1/2}(h,w)e^{-j2\pi\left(\frac{uh2^i}{H}+\frac{vw2^i}{W}\right)}\end{aligned} \tag{1}$$

where $\mathbf{F}_{\mathrm{FFT},i}^{1/2} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2C_i}$ is the feature obtained after applying the FFT to the ith layer's T1 temporal or T2 temporal feature.

After the FFT, the elements of the bitemporal feature $\mathbf{F}_i^{1/2}$ are decomposed into frequency components at each position in $\mathbf{F}_{\mathrm{FFT},i}^{1/2}$. Therefore, a weighting matrix can be used to weight different frequency components to achieve a filtering effect, which can be formulated as

$$\hat{\mathbf{F}}_{\mathrm{FFT},i}^{1/2} = \mathbf{F}_{\mathrm{FFT},i}^{1/2} \otimes \mathbf{M}_i \tag{2}$$

where $\hat{\mathbf{F}}_{\mathrm{FFT},i}^{1/2} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2C_i}$ is the feature after frequency filtering. $\mathbf{M}_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 1}$ represents a weighting mask, and $\otimes$ denotes the elementwise multiplication. It is worth noting that feature maps of different scales are filtered through weight masks corresponding to their resolutions. The method of generating these masks is described in (6) and (7). This strategy optimizes the representation of bitemporal features, thereby suppressing the impact of pseudochange information.

C. Adaptive Frequency Filtering Module

In FIMP, the Fourier feature interaction strategy is implemented by the AFFM. Specifically, after the backbone performs initial feature extraction on the input images, the features from the last four layers are used as the input for the AFFM. Through (1), the bitemporal features are transformed into the frequency domain. Subsequently, the differential frequency-domain feature is obtained by subtraction, which can be formulated as

$$\mathbf{D}_{\mathrm{FFT},i} = \mathbf{F}_{\mathrm{FFT},i}^1 - \mathbf{F}_{\mathrm{FFT},i}^2 \tag{3}$$

where the subtraction of bitemporal features is carried out on an element-by-element basis. The frequency-domain difference features $\mathbf{D}_{\mathrm{FFT},i}$ can provide preliminary guidance for retrieving change information.

Since a simple subtraction is performed on the features from bitemporal images, $\mathbf{D}_{\mathrm{FFT},i}$ contains redundant change information. If used directly to generate an attention map, it may have a negative impact. Therefore, a 1×1 convolution is first used to update different frequency components of the frequency-domain features, thereby optimizing the feature representation. In addition, according to the linear properties of the Fourier transform, the Fourier transform of the difference between the signal S1 and the signal S2 is equal to the difference between the Fourier transform of the two signals. That can be formulated as

$$\mathcal{F}(S_1 - S_2) = \mathcal{F}(S_1) - \mathcal{F}(S_2) \tag{4}$$

where $\mathcal{F}(S_1)$ and $\mathcal{F}(S_2)$, respectively, represent the Fourier transforms of signals S_1 and S_2.

In the frequency domain, a point does not represent a local area of the image but rather information about a specific frequency throughout the entire image [50]. Therefore, the 1×1 convolution can also be considered a preliminary filtering of the original bitemporal image difference features, representing an initial mining of change information. This process can be formulated as

$$\hat{\mathbf{D}}_{\mathrm{FFT},i} = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{D}_{\mathrm{FFT},i}\right)\right)\right) \tag{5}$$

where $\hat{\mathbf{D}}_{\mathrm{FFT},i}$ represents the updated difference features, σ refers to the parametric rectified linear unit (PReLU) used as a nonlinear activation function, and BN stands for the batch normalization.

In order to mine the information carried by frequency components relevant to the change detection task during the frequency-domain interaction process, an attention-based method is employed to adaptively weight different frequency components. In this context, the attention map is generated based on the differential feature $\hat{\mathbf{D}}_{\mathrm{FFT},i}$. This process can be formulated as

$$\hat{\mathbf{F}}_{\mathrm{FFT},i}^{1/2} = \mathbf{F}_{\mathrm{FFT},i}^{1/2} \otimes \mathbf{A_i}\left(\hat{\mathbf{D}}_{\mathrm{FFT},i}\right) \tag{6}$$

where $\mathbf{A_i} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 1}$ represents a learnable attention mask. According to Python's broadcast mechanism, the generated attention mask and feature $\mathbf{F}_{\mathrm{FFT},i}^1$ perform channel-by-channel multiplication, thereby filtering out information irrelevant to the downstream task.

Inspired by the CBAM attention mechanism [51], the mask generation is based on the spatial attention mechanism in CBAM. This approach focuses on identifying the importance

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of various frequency components rather than distinguishing between individual channels. A shared weight mask across all channels ensures consistent attention to the same spatial locations, i.e., the frequency components, across the entire network. The differential features first undergo max pooling and average pooling along the channel dimension. Then, the frequency-domain feature attention map is generated through a 7×7 convolution followed by a Sigmoid function. The aforementioned process can be formulated as

$$\mathbf{A_i} = \delta\left(\mathrm{Conv}_{7\times7}\left(\mathrm{Cat}\left[\mathrm{MaxPool}(\hat{\mathbf{D}}_{\mathrm{FFT},i}), \mathrm{AvgPool}\left(\hat{\mathbf{D}}_{\mathrm{FFT},i}\right)\right]\right)\right) \tag{7}$$

where δ represents the Sigmoid function, MaxPool($\cdot$) refers to the max pooling layer, AvgPool($\cdot$) refers to the average pooling layer, and Cat[$\cdot$] is the concatenation operation along the channel dimension.

According to (3), the obtained attention map is used as weights to multiply with the bitemporal frequency-domain features to achieve filtering. Finally, the filtered bitemporal frequency-domain features $\hat{\mathbf{F}}^{1/2}_{\mathrm{FFT},i}$ are transformed back into the spatial domain through the IFFT to obtain $\tilde{\mathbf{F}}^{1/2}_{i}$ as follows:

$$\tilde{\mathbf{F}}^{1/2}_{i} = \mathcal{F}^{-1}\left(\hat{\mathbf{F}}^{1/2}_{\mathrm{FFT},i}\right) \tag{8}$$

where $\mathcal{F}^{-1}$ represents the IFT operation.

Because numerical changes of individual elements in the frequency domain can affect the global representation of the original features [49], such an approach can mitigate the impact of style differences. Moreover, since this is a differential frequency-domain attention, it places more emphasis on preserving information relevant to the downstream change detection task during the filtering process.

D. Temporal Change Enhancement Module

For change detection tasks, thoroughly interacting and modeling the temporal correlations between bitemporal images are imperative [52]. A common practice was to concatenate or subtract the features of bitemporal images to model the temporal relationship between them, thereby identifying changing targets. However, when dealing with high-resolution remote sensing imagery with complex geographical distributions, such methods exhibit poor robustness and have limited ability to recognize changing targets. This is because such a method struggles to capture subtle spatial changes and the complex dynamic relationships between temporal phases.

To overcome the aforementioned shortcomings, a TCEM is proposed. As shown in Fig. 2, the spatiotemporal dependence between two temporal phases is modeled by mixing the channel dimensions of the bitemporal features. For the bitemporal features after adaptive frequency filtering, feature interaction is first conducted through a mixing block. Specifically, for the bitemporal features $\left\{\tilde{\mathbf{F}}^{1}_{i}, \tilde{\mathbf{F}}^{2}_{i}\right\} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i}$, they are first interlaced along the channel dimension, which can be formulated as

$$\mathbf{F}_{\mathrm{mix},i} = \begin{cases} \tilde{\mathbf{F}}^{1}_{i,[k/2]}, & \text{if } k \text{ is even} \\ \tilde{\mathbf{F}}^{2}_{i,[k/2]}, & \text{if } k \text{ is odd} \end{cases} \tag{9}$$

where $\mathbf{F}_{\mathrm{mix},i} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2C_i}$ represents the mixed features and k is the index of the channel dimension of feature $\mathbf{F}_{\mathrm{mix},i}$. $\tilde{\mathbf{F}}_{i,[k/2]}$ and $\tilde{\mathbf{F}}^{2}_{i,[k/2]}$ denote the 2-D tensors of size $(H/2^i)\times(W/2^i)$ of the $[k/2]$th channel of features $\tilde{\mathbf{F}}^{1}_{i}$ and $\tilde{\mathbf{F}}^{2}_{i}$, respectively. [$\cdot$] denotes the downward rounding operation.

After the feature mixing, every adjacent pair of channels in $\mathbf{F}_{\mathrm{mix},i}$ represents the semantic information of the bitemporal image, respectively. This characteristic of $\mathbf{F}_{\mathrm{mix},i}$ provides the possibility to model the spatiotemporal relationships between the bitemporal images simultaneously. Therefore, the mixing block employs grouped convolution to interact with the bitemporal features under the condition of modeling spatiotemporal relationships. Specifically, for the feature $\mathbf{F}_{\mathrm{mix},i}$, group convolution is performed with a 3×3 convolution kernel, and the number of groups is C_i. During the convolution process, for a feature map with $2C_i$ channels, it is accordingly divided into C groups, meaning that the dimension of each group of features is $(H/2^i) \times (W/2^i) \times 2$. Therefore, each group of convolution kernels can simultaneously perform spatial and temporal convolution on the bitemporal features of the corresponding channels, fully exploiting the morphological representation of the geographical features and the temporal change relationships. The above process can be formulated as follows:

$$\hat{\mathbf{F}}_{\mathrm{mix},i} = \sigma\left(\mathrm{BN}\left(\mathrm{GroupConv}_{3\times3}\left(\mathbf{F}_{\mathrm{mix},i}\right)\right)\right) \tag{10}$$

where σ represents the PReLU activation function, BN stands for batch normalization layer, and $\mathrm{GroupConv}_{3\times3}$ is the grouped convolution layer with a kernel size of 3×3.

To enhance the change information contained in $\hat{\mathbf{F}}_{\mathrm{mix},i}$ and optimize the feature representation, the bitemporal features $\left\{\tilde{\mathbf{F}}^{1}_{i}, \tilde{\mathbf{F}}^{2}_{i}\right\}$ are subtracted to highlight the change information. The differential change information is then conveyed into feature $\hat{\mathbf{F}}_{\mathrm{mix},i}$ in a manner similar to attention weights. This process can be formulated as

$$\mathbf{D}_i = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\left|\tilde{\mathbf{F}}^{1}_{i} - \tilde{\mathbf{F}}^{2}_{i}\right|\right)\right)\right) \tag{11}$$

$$\mathbf{F}_{c,i} = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\hat{\mathbf{F}}_{\mathrm{mix},i} \otimes \mathbf{D}_i \oplus \hat{\mathbf{F}}_{\mathrm{mix},i}\right)\right)\right) \tag{12}$$

where $\mathbf{F}_{c,i} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C'_i}$ represents the output of the ith TCEM and $\oplus$ represents the elementwise addition.

As illustrated in Fig. 2, the TCEM is applied to bitemporal feature pairs at four different scales to capture multiscale change object information and provide preliminary retrieval guidance for subsequent detection tasks.

E. U-Fusion Change Perception Module

After obtaining change features at different scales, how to aggregate their contextual information and guide it into the upsampling process is the key to perceiving multiscale changing geographical features. For the scale features extracted by the encoder at different levels, shallow features contain information such as color and texture, which helps in perceiving change boundaries. Deep features contain rich semantic information, which aids in identifying changing objects. Therefore, the U-fusion change perception module (UCPM) is used to bridge the semantic gap between multiscale

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features, aggregating the global contextual information of the image. The UCPM consists of a multiscale downsampling feature aggregation block (MDFA) and a multiscale upsampling feature aggregation block (MUFA). By employing a U-shaped architecture to bidirectionally aggregate contextual information across different levels, the ability to perceive changing objects at various scales is enhanced.

Specifically, in the multiscale downsampling block, for the features $\mathbf{F}_{c,i} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i'}$ and $\mathbf{F}_{c,i+1} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 2C_i'}$ after the TCEM from two adjacent levels, the feature map sizes are first unified through feature reshaping. As shown in Fig. 2, for $\mathbf{F}_{c,i}$, a reorganized feature map $\hat{\mathbf{F}}_{c,i} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 4C_i'}$ is obtained by sampling every other pixel along the horizontal and vertical directions on a per-channel basis. The information contained in every four adjacent channels in $\hat{\mathbf{F}}_{c,i}$ corresponds to the information of a single channel in $\mathbf{F}_{c,i}$. Therefore, each four adjacent channels in $\hat{\mathbf{F}}_{c,i}$ are convoluted in the form of group convolution, and the original feature representation is aggregated

$$\mathbf{F}'_{c,i} = \sigma\left(\mathrm{BN}\left(\mathrm{GroupConv}_{3\times3}\left(\hat{\mathbf{F}}_{c,i}\right)\right)\right) \tag{13}$$

where $\mathbf{F}'_{c,i} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 2C_i'}$ and the number of groups in the group convolution is C_i'.

The feature reshaping operation to some extent avoids the information loss caused by downsampling operations such as pooling layers and vanilla convolution. Then, after adding the features $\mathbf{F}'_{c,i}$ and $\mathbf{F}_{c,i+1}$ with the same dimension, the context information is preliminarily aggregated through a residual convolution block as

$$\mathbf{F}_a = \mathbf{F}'_{c,i} \oplus \mathbf{F}_{c,i+1} \tag{14}$$

$$\mathbf{F}'_a = \sigma(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{F}_a))) \oplus \mathbf{F}_a \tag{15}$$

where $\mathbf{F}'_a \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 2C_i'}$ represents the preliminarily aggregated feature.

To efficiently capture the dependencies between channels while minimizing the loss in dimensionality, an efficient channel attention (ECA) layer [53] is used to better learn the change semantics contained between channels, thereby optimizing the feature representation of $\mathbf{F}'_a$. ECA employs a nondiminishing, local cross-channel interaction strategy that significantly reduces the complexity of conventional channel attention computations while maintaining the ability to capture important channels.

Finally, the output from the ECA layer serves as the input for the next MDFA block, continuing the feature aggregation and downsampling operations with $\mathbf{F}_{c,i+2}$. In the upsampling section, the structure of the MUFA block is similar to that of the MDFA block, with the only difference being that the feature reorganization downsampling operation is replaced with feature reorganization upsampling. Ultimately, the optimized change features from each layer are fed into a simple decoder and classification head via skip connections to obtain the final change detection prediction results $\mathbf{P} \in \mathbb{R}^{H\times W\times 2}$.

F. Loss Function

For the binary change detection tasks, it can actually be considered a binary classification task. Therefore, the cross-entropy loss function is often used as the loss function for model training, which can be formulated as

$$\mathcal{L}_{\mathrm{ce}}(\mathbf{Y}, \mathbf{P}) = -\frac{1}{H_0 \times W_0} \sum_{i,j} \mathbf{Y}_{i,j} \log \mathbf{P}_{i,j} \tag{16}$$

where $\mathbf{Y}$ is the ground truth change detection map of the input bitemporal images. H_0 and W_0 are the height and width of the bitemporal images, respectively.

IV. Experiments

In this section, three public change detection datasets utilized for the experiments are discussed first. Then the relevant experimental details are introduced. Finally, experimental results comparing FIMP with the state-of-the-art methods and the ablation study results of FIMP are presented.

A. Dataset Introduction

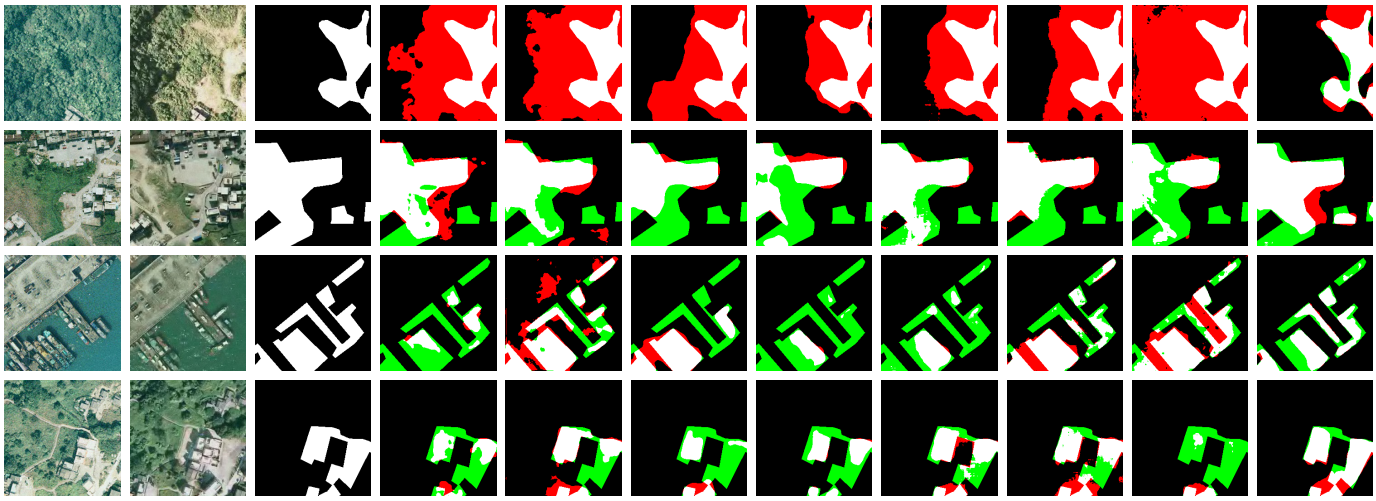
The Sun Yat-Sen University CD Dataset (SYSU-CD) [54], the High-Resolution Complex Urban Scene CD Dataset (HRCUS-CD) [55], and the Wuhan University Building CD Datasets (WHU-CD) [56] were selected for experimental validation. These datasets encompass various types of changes, such as building alterations, vegetation changes, and road construction, allowing for a comprehensive evaluation of the algorithm's performance.

1) Sun Yat-Sen University CD Dataset: This dataset comprises a total of 800 pairs of aerial images with a resolution of 0.5 m. It documents the various changes that occurred in Hong Kong, China, from 2014 to 2017, including maritime construction, vegetation changes, road reconstruction, and so on. The authors crop the original data into nonoverlapping image patches sized 256 × 256 and divide them into training, validation, and test sets in a 6:2:2 ratio. Ultimately, the training, validation, and test sets include 12 000, 4000, and 4000 pairs of images, respectively.

2) High-Resolution Complex Urban Scene CD Dataset: This dataset contains a total of 11 388 pairs of high-resolution remote sensing images with a resolution of 0.5 m, each sized 256 × 256. It documents the changes that occurred in Zhuhai City, China, from 2010 to 2018 and from 2019 to 2022. The dataset primarily focuses on various types of building changes in complex scenes. The authors divide the original data into training, validation, and test sets in a 7:2:1 ratio. Ultimately, the training, validation, and test sets include 7974, 2276, and 1138 pairs of images, respectively.

3) Wuhan University Building CD Datasets: This dataset comprises a single pair of aerial images with a resolution of 0.2 m, each sized 32 207 × 15 354. It documents the changes in Christchurch, New Zealand, from the aftermath of the earthquake in 2011 to the reconstruction in 2012. This dataset primarily focuses on changes in buildings. For the convenience of model training, the original data were cropped into nonoverlapping image patches sized 256 × 256 and randomly divided. Ultimately, the training, validation, and test sets include 5950, 742, and 742 pairs of images, respectively.

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Fig. 4. Qualitative results of the proposed method compared with other methods on the SYSU-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) BIT [18]. (e) DSAMNet [54]. (f) TFI [57]. (g) AFCF3D-Net [58]. (h) P2V-CD [59]. (i) EATDer [60]. (j) MAFGNet [61]. (k) FIMP. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and green pixels represent false negatives.

B. Experimental Details

All experiments use the Pytorch platform, and the model is trained and tested on a workstation using a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. Data augmentation on the training set is performed by randomly flipping and exchanging the order of the bitemporal images. AdamW is used as the optimizer to optimize the model parameters, with a weight decay of 0.01 and an initial learning rate of 0.001. A cosine annealing strategy is employed to adjust the learning rate, with the minimum learning rate set to 0.0001. The training is set for 200 epochs, with a batch size of 32.

Five commonly used metrics in change detection tasks, including the overall accuracy (OA), precision (Pre), recall (Rec), F1 score (F1), and Intersection over Union (IoU), are employed for quantitative analysis [42].

C. Comparison With State-of-the-Art Methods

To verify the effectiveness of FIMP, it is compared with seven methods: BIT [18], DSAMNet [54], TFI [57], AFCF3D-Net [58], P2V-CD [59], EATDer [60], and MAFGNet [61]. All comparative methods were implemented using their respective open-source codes and were tested according to the parameter settings provided in the original articles. Among them, BIT [18], DSAMNet [54], TFI [57], and AFCF3DNet [58] all use ResNet18 [62] as their backbone. P2V-CD [59], EATDer [60], and MAFGNet [61] utilize backbones that are modules specifically proposed in their respective methods. These methods are all representative or state-of-the-art in the field of change detection in recent years. Comparing them effectively shows the superior performance of FIMP.

Among these methods, BIT [18] represents bitemporal images as semantic tokens. It uses a transformer encoder to model the contextual information of the images and refines the feature representation through a transformer decoder to achieve change detection. DSAMNet [54] enhances the encoder's ability to capture change information through deep supervision and integrates convolutional attention blocks into the metric module to refine the change detection results. TFI [57] achieves good detection results in complex terrain distribution scenarios through the interaction between temporal features and a multilevel feature refinement module. AFCF3D-Net [58] utilizes 3-D convolution for feature extraction and fusion, thereby reducing the semantic gap between adjacent multiscale features. P2V-CD [59] extends bitemporal images into video frame sequences to address the issue of insufficient temporal modeling by the model. It fully exploits the geometric information and temporal correlations of images through spatial and temporal convolutional modules. EATDer [60] utilizes change boundary information to guide the transformer decoder in capturing long-distance contextual information, accurately identifying change objects while precisely locating their boundaries. MAFGNet [61] captures global contextual information of images using graph CNNs. It integrates multiscale feature information by fusing features extracted from spatial graph convolutional networks and channel graph convolutional networks.

1) Qualitative Evaluation: The qualitative results on the three datasets are shown in Figs. 4–6. In the prediction maps of various methods, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and green pixels represent false negatives.

From Fig. 4, it can be observed that due to the uncertainty of imaging conditions, the color, brightness, and other style representations of bitemporal images in SYSU-CD are significantly inconsistent. Among the seven comparison methods, AFCF3D-Net and P2V-CD achieved relatively better detection results. This may be attributed to the integration of spatial and temporal features in these two methods, which enhances the network's ability to capture change regions. However, both methods still exhibit a higher number of false positives in pseudochange areas. As seen from the first row, FIMP effectively mitigates the impact of pseudochanges caused by inconsistencies in vegetation phenotypes, especially in cases where other methods produce a large number of false positives.

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This might be due to the strategy of frequency interaction, which captures more precise change semantics while mitigating style differences. From the visualization results of the second to fourth rows, it can be observed that the proposed method is better at extracting morphological information of changing objects and is also capable of identifying smaller detection targets when facing multiscale changes. This might be because FIMP, through its U-fusion approach, bridges the semantic gap between feature maps of different scales while fully aggregating global context information, thereby reducing the missed and false detections of change areas.

From Fig. 5, it can be observed that compared to SYSU-CD, the changing objects in HRCUS-CD exhibit dense and small characteristics, and the change scenarios are more complex, making accurate identification of terrain changes even more challenging.

When addressing small and densely clustered change regions, P2V-CD demonstrates fewer false detections compared to other methods but suffers from a higher rate of missed detections. This is likely due to its robust modeling of temporal correlations, allowing for more accurate capture of change semantics, though it lacks sensitivity to multiscale objects. TFI, AFCF3D-Net, and EATDer detect more complete change regions, but they also produce excessive false positives, likely because they aggregate multiscale change features to some extent. Compared to these methods, FIMP achieves a better balance between false positives and missed detections, resulting in superior detection performance overall. Furthermore, from the results of the second and third rows, it can be seen that when facing pseudochange effects such as building shadows caused by the uncertainty of imaging conditions, FIMP is able to more meticulously delineate the true change boundaries in pseudochange areas compared to other methods. This may be because the Fourier feature interaction strategy effectively filters out some pseudochanges, and TCEM fully simulates the temporal correlation of the bitemporal image.

WHU-CD is a dataset focused solely on building changes. As can be seen from Fig. 6, the morphological features of buildings are diverse, which poses a challenge to accurately depicting the boundaries of changed buildings. Compared to other methods, FIMP can more accurately define change boundaries, reducing false detections while more completely recognizing change areas. FIMP achieves better detection results by deeply interacting between bitemporal images and fully aggregating multiscale features, thereby narrowing the semantic gap while aggregating contextual information.

2) Quantitative Evaluation: The quantitative results on the three experimental datasets are shown in Table I. In these tables, red indicates the highest scores for each quantitative metric, while blue denotes the second highest scores.

It can be observed that FIMP achieves the best results on the three datasets. As an example of a more comprehensive evaluation metric, on SYSU-CD, HRCUS-CD, and WHU-CD, compared to the second highest F1 score, FIMP improved by 0.32%, 2.67%, and 1.73%, respectively. Compared to the second highest IoU, FIMP improved by 0.47%, 3.48%, and 3.09%, respectively. For BIT and MAFGNet, although they capture long-range semantic information in different ways, they neglect the effective utilization of change features at different levels. Therefore, their detection performance is generally mediocre when facing complex scenes. For DSAMNet, P2V-CD, and EATDer, although they can extract more representative change features, they merely perform a simple fusion of multilevel features, ignoring the impact of the semantic gap between deep and shallow features. Among all comparison methods, TFI and AFCD3D-Net also achieve good results on various datasets. This might be because they also fully aggregate multilevel features of different scales, capturing more contextual information while narrowing the semantic gap between deep and shallow features. However, these two methods do not consider the impact of style differences between bitemporal images and directly interact with the extracted bitemporal features. Compared to other methods, FIMP uses frequency interaction, employing adaptive frequency filtering to mine information relevant to the change detection task and adjust the global representation of bitemporal features, thereby mitigating the impact of style differences between bitemporal images to some extent.

D. Computational Complexity Analysis

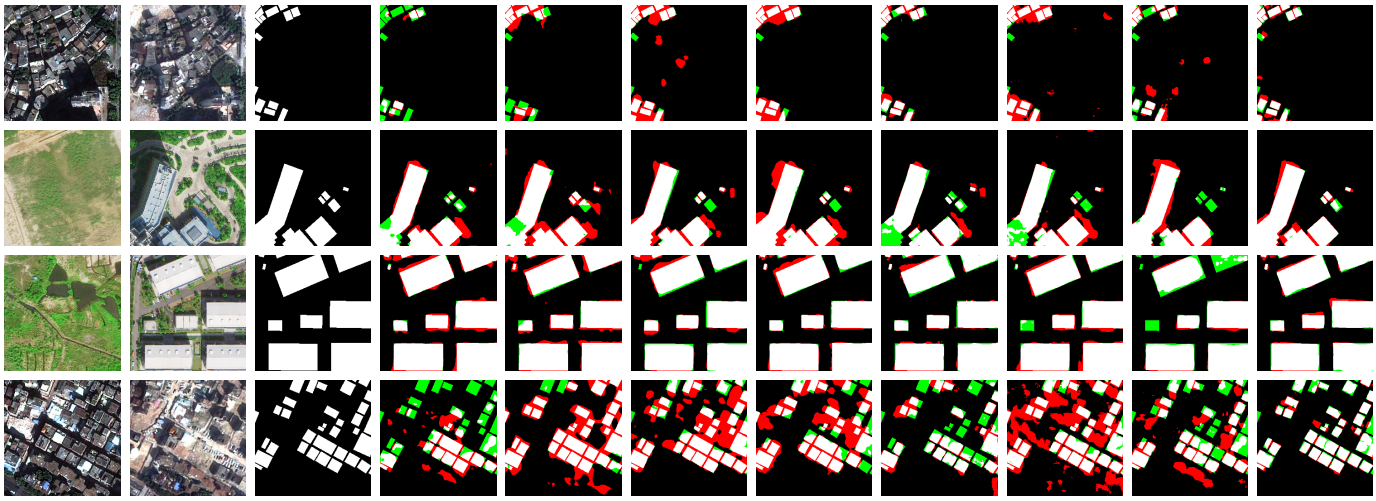
To further evaluate the model size and computational complexity, the comparison results of the proposed method with the other seven comparison methods in terms of FLOPs and the number of parameters are shown in Table II. The best results for each metric are highlighted in bold in Table II. w/o indicates that no pretrained backbone is used but is replaced by carefully designed modules in their article. It can be observed that FIMP achieves better detection results with the smallest number of parameters and the least computational complexity, further validating the performance of the proposed method.

E. Ablation Studies and Parameter Analysis

To verify the effectiveness of each module in FIMP, ablation studies were conducted on FIMP across three datasets under the same experimental conditions. The number of layers in UCPM is set to 3 by default. In the baseline model, the TCEM is replaced with a concatenation operation and a single 1×1 convolution layer that includes batch normalization and a PReLU activation function. The quantitative results of the ablation study in terms of F1 score and IoU are summarized in Table III.

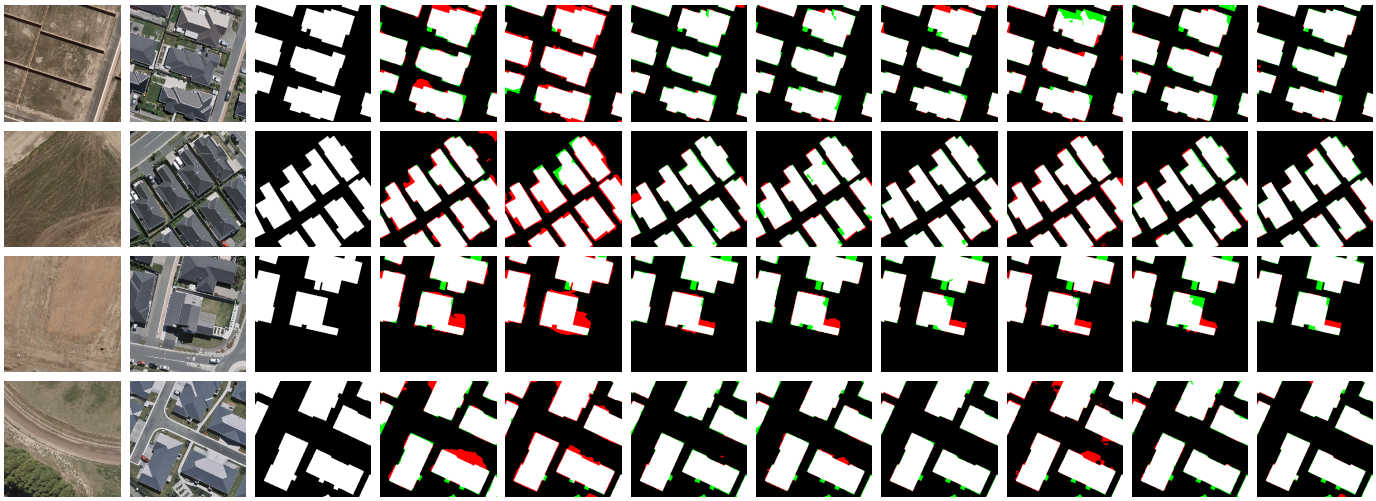
1) Effectiveness of Fourier Feature Interaction Strategy: The Fourier feature interaction strategy optimizes the bitemporal feature representation by transforming the original features into the frequency domain and adaptively weighting different frequency components. This approach effectively mitigates the impact of style differences while mining change information. From the first and second rows of Table III, it can be seen that when only the AFFM is added to the baseline model, the F1 scores on the three datasets increase by 1.24%, 0.33%, and 0.26%, respectively, and the IoU increase by 2.65%, 0.42%, and 0.46%, respectively. Furthermore, when only removing the AFFM from the complete FIMP, the F1 score and IoU on the three datasets decrease by 0.83%, 1.23%, 0.39% and 1.21%, 1.62%, 0.70%, respectively.

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Fig. 5. Qualitative results of the proposed method compared with other methods on the HRCUS-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) BIT [18]. (e) DSAMNet [54]. (f) TFI [57]. (g) AFCF3D-Net [58]. (h) P2V-CD [59]. (i) EATDer [60]. (j) MAFGNet [61]. (k) FIMP. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and green pixels represent false negatives.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 6. Qualitative results of the proposed method compared with other methods on the WHU-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) BIT [18]. (e) DSAMNet [54]. (f) TFI [57]. (g) AFCF3D-Net [58]. (h) P2V-CD [59]. (i) EATDer [60]. (j) MAFGNet [61]. (k) FIMP. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and green pixels represent false negatives.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT CHANGE DETECTION METHODS ON SYSU-CD, HRCUS-CD, AND WHU-CD DATASETS, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND BEST RESULTS ARE BLUE. ALL RESULTS OF THE THREE EVALUATION METRICS ARE DESCRIBED AS PERCENTAGES (%)

Method	SYSU-CD					HRCUS-CD					WHU-CD				
	OA	F1 Score	IoU	Pre	Rec	OA	F1 Score	IoU	Pre	Rec	OA	F1 Score	IoU	Pre	Rec
BIT [18]	89.63	77.93	63.84	78.25	77.61	98.86	67.54	50.99	72.31	63.35	98.85	87.31	77.47	85.36	89.35
DSAMNet [54]	87.60	75.59	60.76	70.57	81.39	98.74	66.22	49.50	66.66	65.78	98.48	84.40	73.01	77.56	92.56
TFI [57]	92.41	83.68	71.94	84.83	82.56	99.06	75.00	60.01	74.58	75.43	99.42	93.44	87.68	93.44	93.01
AFCF3D-Net [58]	92.04	82.95	70.87	83.76	82.16	98.99	74.01	58.74	72.06	76.07	99.42	93.40	87.62	94.58	92.25
P2V-CD [59]	90.36	78.86	65.10	81.66	76.25	98.99	71.16	55.23	76.85	66.25	99.44	93.56	87.90	95.18	92.00
EATDer [60]	91.03	80.96	68.01	81.02	80.90	98.62	67.25	50.66	60.56	75.61	99.03	89.27	80.62	87.85	90.74
MAFGNet [61]	91.07	80.69	67.62	82.36	79.08	98.80	63.97	47.02	73.19	56.81	99.29	91.99	85.17	91.87	92.12
FIMP	92.69	84.00	72.41	86.78	81.38	99.19	77.67	63.49	80.49	75.04	99.58	95.29	90.99	96.88	93.74

To further demonstrate the effectiveness of the AFFM, class activation mapping (CAM) is utilized to visualize the bitemporal feature maps before and after filtering. CAM intuitively demonstrates through heatmaps how the model perceives certain areas to be highly relevant to specific categories. The visualization results are displayed in Fig. 7. In the visualized feature maps, areas that the model considers to be more related to the change category appear redder, while areas with lower

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TABLE II

COMPARISON OF FLOPS AND PARAMETERS BETWEEN THE PROPOSED METHOD AND SEVEN COMPARISON METHODS

Method	Backbone	Params (M)	FLOPs (G)
BIT [18]	Resnet18	3.04	10.90
DSAMNet [54]	Resnet18	16.95	72.31
TFI [57]	Resnet18	28.37	9.67
AFCF3D-Net [58]	Resnet18	17.64	31.71
P2V-CD [59]	w/o	5.42	33.03
EATDer [60]	w/o	6.59	23.46
MAFGNet [61]	w/o	6.58	62.18
FIMP	EfficientNet-b4	**2.45**	**2.85**

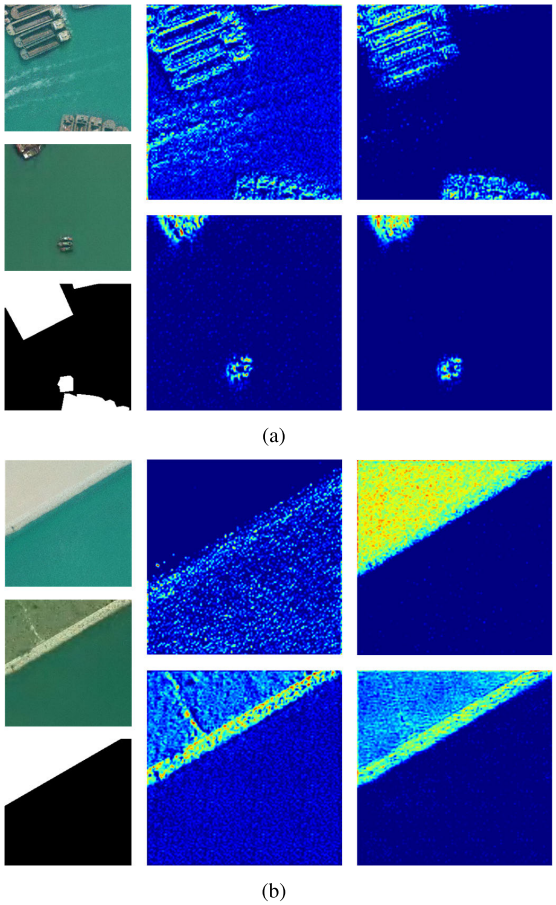


Fig. 7. Visualization results of feature maps on the SYSU-CD dataset (for the change category). Red and blue, respectively, represent areas that are highly relevant and lowly relevant to the change category. In (a) and (b), the first column from top to bottom represents the T1 image, the T2 image, and the ground truth; the second column represents the bitemporal feature maps before frequency filtering; and the third column represents the feature maps after bitemporal frequency filtering.

relevance to the change category appear bluer. From the first and second columns of Fig. 7(a), it can be observed that before processing with AFFM, the wakes of ships on the sea or waves, as well as nonchanging areas with inconsistent styles in bitemporal images, are identified by the network as regions related to the change category. From the first and second columns of Fig. 7(b), it can be seen that there is a significant color inconsistency in the sea surface part of the bitemporal images, causing the encoder to mistakenly identify the sea surface as a change area, which will severely affect the training of the model. However, after passing through

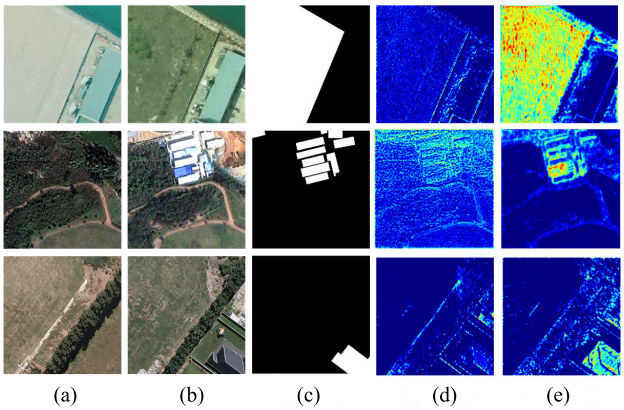


Fig. 8. Partial visualization results of feature maps from the ablation study of TCEM on three datasets. (Top to bottom) Three rows are from the SYSU-CD, HRCUS-CD, and WHU-CD datasets, respectively. (d) Results of replacing TCEM with concatenation and 1 × 1 convolution. (a) T1 image. (b) T2 image. (c) Label. (d) Concat. (e) TCEM.

the AFFM module, the aforementioned pseudochanges are effectively filtered out by AFFM. At the same time, the model's attention to the actual change areas is increased. This further shows the effectiveness of mitigating the impact of pseudochanges such as style differences through AFFM.

2) Effectiveness of the TCEM: TCEM fully interacts with the bitemporal features optimized after filtering to model temporal correlations and highlight change areas. From the first and third rows of Table III, it can be seen that when only TCEM is added to the baseline model, the F1 scores on the three datasets increase by 1.95%, 0.66%, and 0.36%, respectively, and the IoU scores increase by 2.78%, 0.84%, and 0.65%, respectively. From the last and third-to-last rows of Table III, it can be observed that when TCEM is solely removed from the complete FIMP, the F1 scores and IoU on the three datasets decrease by 1.05%, 1.36%, 0.35% and 1.53%, 1.80%, 0.62%, respectively.

From Fig. 8, it can be seen that the concatenation operation has a poor recognition effect on change areas across the three datasets. This is because simple concatenation merely fuses the bitemporal features together, making it difficult to model temporal correlations. However, TCEM enhances the recognition of change areas by utilizing the difference between bitemporal features, under the premise of fully modeling temporal correlations, resulting in improved recognition effects.

In addition, various feature mixing methods have been explored in greater detail. As shown in Table IV, $\mathbf{A} - \mathbf{B}$ represents the subtraction of bitemporal features, $|\mathbf{A} - \mathbf{B}|$ indicates the absolute value of the subtracted features, Concat refers to the concatenation of bitemporal features along the channel dimension, and Add signifies the addition of the bitemporal features. The best results are highlighted in bold. Experimental results indicate that while these simple feature fusion methods offer advantages for different datasets, none of them fully capture the semantic relationships between the bitemporal features. Among these methods, TCEM achieved the highest F1 score, indicating that it effectively models the temporal correlations between bitemporal features.

3) Effectiveness of the UCPM: UCPM enhances the model's ability to capture change objects of different scales by

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TABLE III
ABLATION STUDY RESULTS OF AFFM, TCEM, AND UCPM IN FIMP ON THREE DIFFERENT DATASETS

AFFM	TCEM	UCPM	SYSU-CD		HRCUS-CD		WHU-CD	
			F1 Score	IoU	F1 Score	IoU	F1 Score	IoU
×	×	×	0.8054	0.6742	0.7504	0.6006	0.9440	0.8940
✓	×	×	0.8170	0.6907	0.7537	0.6048	0.9466	0.8986
×	✓	×	0.8249	0.7020	0.7570	0.6090	0.9476	0.9005
×	×	✓	0.8250	0.7022	0.7567	0.6087	0.9478	0.9008
✓	✓	×	0.8272	0.7054	0.7658	0.6206	0.9509	0.9065
✓	×	✓	0.8295	0.7088	0.7631	0.6169	0.9494	0.9037
×	✓	✓	0.8317	0.7120	0.7644	0.6187	0.9490	0.9029
✓	✓	✓	**0.8400**	**0.7241**	**0.7767**	**0.6349**	**0.9529**	**0.9099**

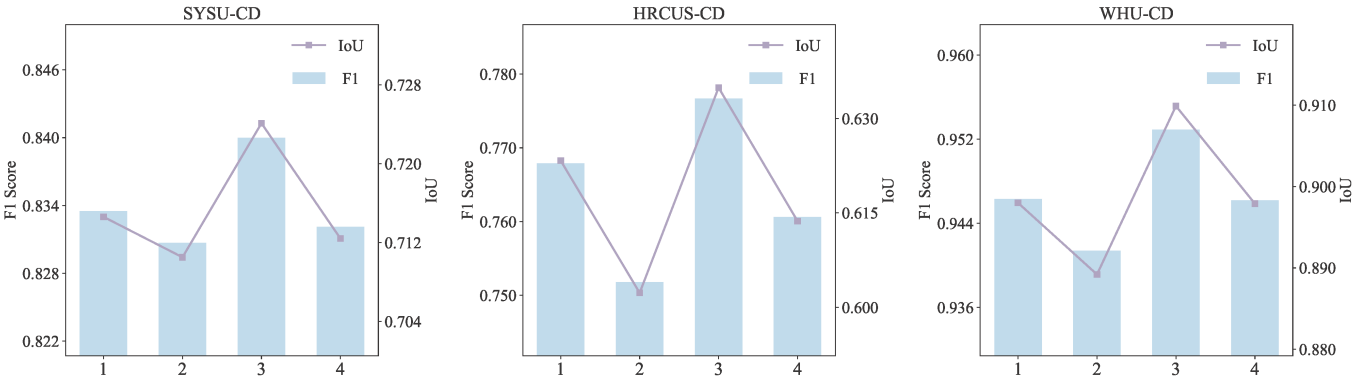


Fig. 9. Experimental results of UCPM with different numbers (1, 2, 3, and 4) of layers on the three datasets. The blue bars represent the F1 score, and the purple line represents the IoU.

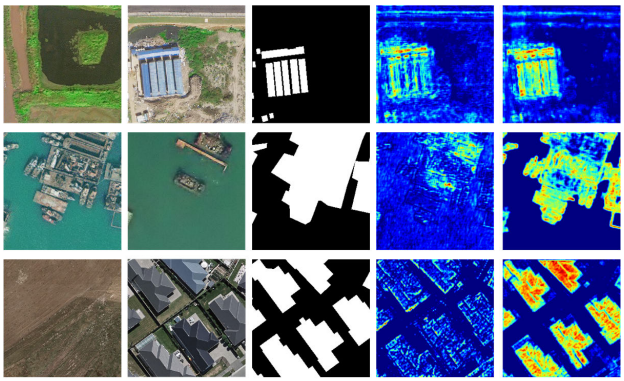


Fig. 10. Partial visualization results of feature maps from the ablation study of UCPM on three datasets. (Top to bottom) Three rows are from the SYSU-CD, HRCUS-CD, and WHU-CD datasets, respectively. (a) T1 image. (b) T2 image. (c) Label. (d) Before UCPM. (e) After UCPM.

TABLE IV
ABLATION EXPERIMENTS OF FEATURE MIXING METHODS. THE BEST RESULTS ARE HIGHLIGHTED IN **BLOD** (UNDER F1 SCORE)

Mixing Method	SYSU-CD	HRCUS-CD	WHU-CD
$\mathbf{A}-\mathbf{B}$	83.27	75.57	95.09
$\lvert\mathbf{A}-\mathbf{B}\rvert$	82.78	76.38	94.48
Concat	82.95	76.31	94.93
Add	83.21	77.12	94.94
TCEM	**84.00**	**77.67**	**95.29**

bidirectionally aggregating features of various levels through a U-shaped architecture. To verify the effectiveness of UPCM, when it is added to the baseline model individually, it achieved F1 scores of 82.50%, 75.67%, and 94.78% and IoU of 70.22%, 60.87%, and 90.08% on the three datasets, respectively. Compared to the baseline network, the F1 scores increased by 2.04%, 0.63%, and 0.38%, and the IoU increased by 2.82%, 0.81%, and 0.68%, respectively. When only UCPM is removed, the F1 scores and IoUs on the three datasets decrease by 1.28%, 1.09%, and 0.20% and 1.87%, 1.43%, and 0.34%, respectively.

To investigate the effect of UCPM layer count on detection performance, experiments were conducted with one, two, three, and four layers across three datasets, with F1 score and IoU employed as quantitative evaluation metrics. As depicted in Fig. 9, the detection performance exhibits an increasing trend as the number of UCPM layers grows, reaching its peak with three layers. When fewer layers are utilized, the UCPM's capacity for multiscale feature fusion is constrained, limiting its effectiveness in enhancing change detection. Conversely, an excessive number of layers lead to a decline in performance, likely due to overfitting. Therefore, three UCPM layers provide an optimal balance between performance and computational efficiency in the overall model.

From Fig. 10, it can be seen that UCPM effectively aggregates the contextual information contained in multiscale features, enhancing the model's ability to perceive various types of change objects. After processing with UCPM, the model is not only able to recognize change objects of different scales more completely but also accurately delineate the boundaries of changes.

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TABLE V
ABLATION EXPERIMENTS OF DIFFERENT SCALES IN UCPM. THE BEST RESULTS ARE HIGHLIGHTED IN **BLOD** (UNDER F1 SCORE)

Case	SYSU-CD	HRCUS-CD	WHU-CD
Case1	82.70	74.28	94.45
Case2	83.00	76.40	94.51
Case3	82.99	76.22	94.76
Case4	**84.00**	**77.67**	**95.29**

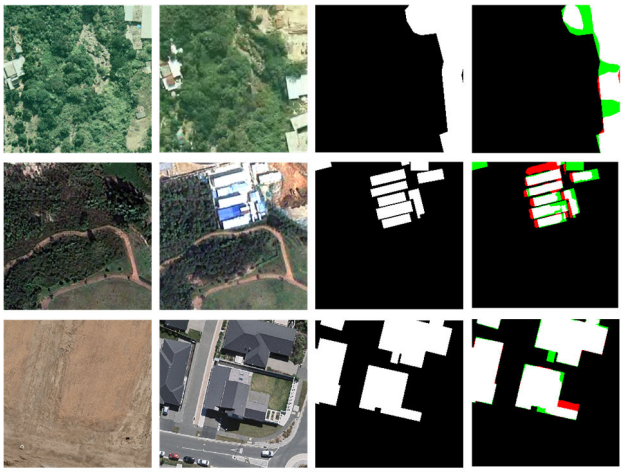


Fig. 11. Failure examples obtained by FIMP on three datasets. (Top to bottom) Three rows are from the SYSU-CD, HRCUS-CD, and WHU-CD datasets, respectively. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and green pixels represent false negatives. (a) T1 image. (b) T2 image. (c) Label. (d) Prediction.

V. DISCUSSION

To further investigate the effectiveness of aggregating multiscale features in UCPM, we designed experiments under the following four cases.

Case1: UCPM is not utilized, and only the deepest layer features are progressively upsampled to generate the prediction results.
Case2: Only the deepest layer MDFA and MUFA within UCPM are used.
Case3: Only the second- and third-layer MDFA and MUFA within UCPM are employed.
Case4: The complete UCPM is applied.

As presented in Table V, the F1 score is the lowest without the use of UCPM (Case1). When UCPM is applied to aggregate partial scale features (Case2 and Case3), the F1 score shows a certain degree of improvement. This could be attributed to the fact that deep semantic features alone are insufficient to accurately capture change details across different scales. The F1 score for Case4 is the highest, further indicating that UCPM effectively bridges the semantic gap between multiscale features through multilayer bidirectional aggregation, thereby enhancing the network's ability to capture complex change objects.

In addition, we also conducted an analysis of the limitations of the proposed method. Fig. 11 illustrates three failure cases of FIMP on the SYSU-CD, HRCUS-CD, and WHU-CD datasets. For the example in the first row of Fig. 11, FIMP has identified most of the change areas, but there are still incomplete detections, with the missing regions highlighted in green. This may be due to the change from grassland to trees in the bitemporal images, where the features of both are quite similar, making it difficult for the model to accurately distinguish them. For the HRCUS-CD and WHU-CD datasets, which primarily focus on building changes, the examples in the second and third rows show that FIMP can relatively accurately locate the change areas and outline the change boundaries. However, due to the similarities between buildings and the ground in some features, the model still exhibits some false positives and false negatives in certain areas. In our opinion, incorporating background feature learning to enhance the model's ability to handle complex backgrounds could be a viable solution. In future work, we will further explore this direction.

VI. CONCLUSION

In this article, a change detection method for high-resolution remote sensing images based on FIMP. Initially, FIMP uses the Fourier feature interaction strategy to enhance bitemporal feature representation. Through the AFFM, it effectively reduces the impact of pseudochanges such as style discrepancies caused by varying imaging conditions. Subsequently, a TCEM is used to model the temporal correlations between bitemporal features, which captures change information while highlighting change areas. Moreover, for the rich semantic information contained in feature maps of different scales, a UCPM is employed. It aggregates contextual information between multiscale change features while narrowing the semantic gap between features at different levels. Conclusively, experiments on three publicly available change detection datasets show that FIMP surpasses the seven most advanced change detection methods. Future research will further focus on weakly supervised change detection methods.

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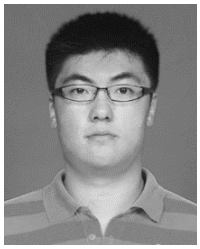
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